

Mathematical Derivations of Diagnostic Thresholds

Five-Step IC Validation Framework

1 Introduction

This appendix provides the rigorous mathematical foundations for the numerical thresholds employed throughout the Five-Step IC Validation Diagnostic Framework. Each threshold is derived from first principles using classical statistical theory—Fisher z-transformation, binomial tests, analysis of variance, and signal-detection theory. Throughout, we denote by ρ (or IC) the Spearman rank correlation between predicted and realized returns.

The framework operates at two complementary resolutions:

1. **Pooled (cross-sectional + time-series)**: a single Spearman correlation computed over all stock-day pairs in a window.
2. **Per-stock**: a separate IC computed for each stock over time, then aggregated.

The thresholds differ accordingly.

2 Threshold 1: $|IC| < 0.02$ as Low-Predictability Bound

2.1 Fisher z-Transformation

For a Spearman correlation r computed from N independent paired observations, the Fisher z-transformation

$$z = \operatorname{arctanh}(r) = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right) \quad (1)$$

is approximately normally distributed with

$$z \sim \mathcal{N} \left(\operatorname{arctanh}(\rho), \frac{1}{N-3} \right), \quad \operatorname{SE}(z) = \frac{1}{\sqrt{N-3}}. \quad (2)$$

This transformation stabilizes the variance and provides a basis for hypothesis testing and confidence intervals.

2.2 Minimum Detectable IC at $\alpha = 0.05$

For a two-sided test of $H_0 : \rho = 0$ against $H_1 : \rho \neq 0$ at significance level α ,

$$|z_{\text{obs}}| > z_{1-\alpha/2} \cdot \operatorname{SE}(z) \implies |r| > \tanh \left(z_{1-\alpha/2} \cdot \frac{1}{\sqrt{N-3}} \right). \quad (3)$$

With $N = 750$ (the approximate number of trading days in the walk-forward validation set; this is the effective sample size treating each day's cross-section as roughly independent [1]):

$$\operatorname{SE}(z) = \frac{1}{\sqrt{750-3}} = \frac{1}{\sqrt{747}} \approx 0.03659, \quad (4)$$

$$r_{\text{crit}}^{(2\text{-sided})} = \tanh(1.96 \times 0.03659) = \tanh(0.07172) \approx 0.0716. \quad (5)$$

For a one-sided test ($H_1 : \rho > 0$) at $\alpha = 0.05$:

$$r_{\text{crit}}^{(1\text{-sided})} = \tanh(1.645 \times 0.03659) = \tanh(0.06019) \approx 0.0601. \quad (6)$$

2.3 Power Analysis

The minimum detectable effect for a given power $1 - \beta$ satisfies

$$z_{\text{eff}} = (z_{1-\alpha/2} + z_{1-\beta}) \cdot \text{SE}(z), \quad r_{\text{min}} = \tanh(z_{\text{eff}}). \quad (7)$$

For 80% power ($z_{1-\beta} = 0.8416$) and $\alpha = 0.05$ (two-sided):

$$z_{\text{eff}} = (1.9600 + 0.8416) \times 0.03659 = 0.1025, \quad (8)$$

$$r_{\text{min}} = \tanh(0.1025) \approx 0.1021. \quad (9)$$

Proposition 2.1. With $N = 750$ observations, the minimum IC detectable at $\alpha = 0.05$ with 80% power is $r_{\text{min}} \approx 0.10$. An IC of 0.02 is therefore far below the detection threshold.

2.4 Why $|\text{IC}| < 0.02$ Constitutes “Low Predictability”

At $r = 0.02$ with $N = 750$,

$$z_{\text{obs}} = \text{arctanh}(0.02) \approx 0.0200, \quad \frac{z_{\text{obs}}}{\text{SE}(z)} = \frac{0.0200}{0.03659} \approx 0.547. \quad (10)$$

The observed correlation is only 0.55 standard errors from zero. The p -value for a two-sided test is $p \approx 0.58$; we would need

$$N_{\text{req}} = \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{r} \right)^2 + 3 \approx \left(\frac{2.8016}{0.02} \right)^2 + 3 \approx 15\,463 \quad (11)$$

observations to detect $r = 0.02$ with 80% power—roughly 20 times the available sample (see Table 1).

Table 1: Power to detect various IC values at $\alpha = 0.05$ with $N = 750$.

IC r	z	$z/\text{SE}(z)$	p -value (one-sided)	Power
0.01	0.0100	0.27	0.39	0.06
0.02	0.0200	0.55	0.29	0.08
0.03	0.0300	0.82	0.21	0.12
0.05	0.0500	1.37	0.09	0.27
0.07	0.0701	1.92	0.03	0.49
0.10	0.1003	2.74	0.003	0.80

The threshold $|\text{IC}| < 0.02$ therefore identifies the regime where the signal is *indistinguishable from noise* given the sample size at hand—a necessary but not sufficient condition for economic significance.

Small- r approximation. For $|r| \ll 1$, $\text{arctanh}(r) \approx r$, so

$$\frac{r}{\text{SE}(z)} \approx r\sqrt{N-3}, \quad r_{\text{crit}} \approx \frac{z_{\alpha/2}}{\sqrt{N-3}}. \quad (12)$$

This yields $r_{\text{crit}} \approx 0.072$ at $\alpha = 0.05$ (two-sided), matching the exact computation.

3 Threshold 2: Held-Out Test IC Gap

3.1 Definition

Let IC_{train} and IC_{held} be the Spearman correlations computed on the training set (first 70% of days) and held-out test set (last 30% of days), respectively. The gap is

$$\Delta = IC_{\text{train}} - IC_{\text{held}}. \quad (13)$$

Under the null hypothesis of no overfitting, $\mathbb{E}[\Delta] = 0$ and both quantities estimate the same population correlation ρ .

3.2 Sampling Noise via Fisher z-Transformation

Using (2), the variance of each IC estimate is $\text{Var}(IC) \approx 1/(N_{\text{eff}} - 3)$, where N_{eff} is the effective number of independent paired observations. For the pooled cross-sectional-time-series Spearman, treating days as the effective unit due to cross-sectional dependence [2],¹ we set N_{eff} to the number of time periods:

$$N_{\text{train}}^{(\text{eff})} \approx 1850 \text{ days}, \quad \text{SE}(IC_{\text{train}}) \approx \frac{1}{\sqrt{1850 - 3}} \approx 0.0233, \quad (14)$$

$$N_{\text{held}}^{(\text{eff})} \approx 795 \text{ days}, \quad \text{SE}(IC_{\text{held}}) \approx \frac{1}{\sqrt{795 - 3}} \approx 0.0355. \quad (15)$$

Proposition 3.1. Under H_0 , the gap Δ has

$$\text{Var}(\Delta) = \text{Var}(IC_{\text{train}}) + \text{Var}(IC_{\text{held}}) - 2 \text{Cov}(IC_{\text{train}}, IC_{\text{held}}). \quad (16)$$

Treating training and held-out sets as independent disjoint samples, $\text{Cov}(\cdot, \cdot) \approx 0$, giving

$$\text{SE}(\Delta) \approx \sqrt{\frac{1}{N_{\text{train}} - 3} + \frac{1}{N_{\text{held}} - 3}}. \quad (17)$$

Numerically:

$$\text{SE}(\Delta) \approx \sqrt{\frac{1}{1847} + \frac{1}{792}} \approx \sqrt{5.41 \times 10^{-4} + 1.26 \times 10^{-3}} \approx \sqrt{1.801 \times 10^{-3}} \approx 0.0424. \quad (18)$$

3.3 Threshold Derivation

The 95% confidence interval width for Δ under H_0 is

$$CI_{95}(\Delta) = \pm z_{0.025} \times \text{SE}(\Delta) \approx \pm 1.96 \times 0.0424 \approx \pm 0.0832. \quad (19)$$

However, a more careful analysis accounts for the fact that IC_{train} and IC_{held} are positively correlated (they are computed from overlapping stock universes with similar features), which *reduces* $\text{Var}(\Delta)$. A conservative lower bound for the effective SE is obtained by assuming the effective sample size per IC is the number of stock-day pairs after filtering (approximately 880 000 for training and 375 000 for held-out in a typical run):

$$\text{SE}(\Delta)_{\min} \approx \sqrt{\frac{1}{880\,000} + \frac{1}{375\,000}} \approx \sqrt{1.14 \times 10^{-6} + 2.67 \times 10^{-6}} \approx 0.00195, \quad (20)$$

$$CI_{95}(\Delta)_{\min} \approx \pm 1.96 \times 0.00195 \approx \pm 0.00382. \quad (21)$$

¹Cross-sectional dependence across stocks inflates the effective sample size adjustment; the conventional approach in factor research is to treat each time period as one observation for variance estimation.

Thus the true $\text{SE}(\Delta)$ lies between 0.002 and 0.042. The chosen thresholds must accommodate both the optimistic and conservative variance estimates:

$$\Delta < 0.01 \quad (\text{healthy} / \text{within noise}), \quad (22)$$

$$0.01 \leq \Delta \leq 0.05 \quad (\text{suspicious}), \quad (23)$$

$$\Delta > 0.05 \quad (\text{pathological} / \text{clear overfitting}). \quad (24)$$

Table 2: Interpretation of the IC gap.

Range	$\Delta / \text{SE}_{\max}$	$\Delta / \text{SE}_{\min}$	Meaning
< 0.01	< 0.24	< 5.1	Within sampling noise (conservative estimate)
$0.01-0.05$	$0.24-1.18$	$5.1-25.6$	Exceeds conservative noise; may indicate mild overfitting
> 0.05	> 1.18	> 25.6	Far exceeds all noise estimates; pathological overfitting

The gap threshold $\Delta = 0.03$ (used in the code as a binary pass/fail) lies at the midpoint of the suspicious range and represents a conservative bound: at $\Delta = 0.03$, even the least optimistic $\text{SE}(\Delta)$ estimate gives a z -ratio of $0.03/0.0424 \approx 0.71$, which is within the 95% CI.

3.4 Connection to Per-Window Gap

In the walk-forward analysis (Step 3), the train/validation gap per window uses analogous reasoning. With $N_{\text{train}} \approx 1008$ days and $N_{\text{val}} \approx 252$ days per window:

$$\text{SE}(\text{IC}_{\text{train}}^{(w)}) \approx 1/\sqrt{1005} \approx 0.0315, \quad (25)$$

$$\text{SE}(\text{IC}_{\text{val}}^{(w)}) \approx 1/\sqrt{249} \approx 0.0634, \quad (26)$$

$$\text{SE}(\Delta^{(w)}) \approx \sqrt{0.0315^2 + 0.0634^2} \approx 0.0708, \quad (27)$$

$$\text{CI}_{95}(\Delta^{(w)}) \approx \pm 0.139. \quad (28)$$

The threshold $\Delta^{(w)} < 0.03$ used in the code ensures that the mean gap across windows does not exceed 0.42 standard errors, well within the noise band.

4 Threshold 3: Coefficient of Variation of IC Across Windows

4.1 Definition

For a set of W walk-forward window ICs $\{\text{IC}_1, \dots, \text{IC}_W\}$,

$$\text{CV}(\text{IC}) = \frac{\sigma_{\text{IC}}}{|\bar{\text{IC}}|}, \quad \bar{\text{IC}} = \frac{1}{W} \sum_{w=1}^W \text{IC}_w, \quad \sigma_{\text{IC}}^2 = \frac{1}{W-1} \sum_{w=1}^W (\text{IC}_w - \bar{\text{IC}})^2. \quad (29)$$

The CV measures the *relative dispersion* of IC across market regimes: a low CV indicates a consistently positive signal, while a high CV indicates regime-dependent predictive performance.

4.2 Expected CV Under the Null ($\text{IC} \approx 0$)

When the true IC is zero, the observed ICs arise purely from estimation noise. For window w with N_w independent observations,

$$\text{IC}_w \sim \mathcal{N}\left(0, \frac{1}{N_w - 3}\right) \quad (\text{first-order approximation via Fisher } z). \quad (30)$$

If all windows have equal size $N_w \approx N$,

$$\text{IC}_w \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_0^2), \quad \sigma_0^2 = \frac{1}{N-3}. \quad (31)$$

The sample standard deviation $s = \sqrt{\frac{1}{W-1} \sum (\text{IC}_w - \bar{\text{IC}})^2}$ satisfies

$$\frac{(W-1)s^2}{\sigma_0^2} \sim \chi_{W-1}^2, \quad \mathbb{E}[s] = \sigma_0 \cdot \sqrt{\frac{2}{W-1}} \frac{\Gamma(W/2)}{\Gamma((W-1)/2)}. \quad (32)$$

For moderate W , $\mathbb{E}[s] \approx \sigma_0$. The expected CV under the null is therefore

$$\mathbb{E}[\text{CV} \mid \rho = 0] \approx \frac{\sigma_0}{|\bar{\text{IC}}|}, \quad (33)$$

and since $\bar{\text{IC}} \rightarrow 0$ in probability, $\text{CV} \rightarrow \infty$. In finite samples, $\bar{\text{IC}}$ is $O_p(1/\sqrt{W})$, so

$$\mathbb{E}[\text{CV} \mid \rho = 0] \approx \frac{\sigma_0}{c/\sqrt{W}} = \frac{\sqrt{W}}{c\sqrt{N-3}} = O\left(\sqrt{\frac{W}{N}}\right), \quad (34)$$

where c is a constant determined by the sampling distribution of $\bar{\text{IC}}$. For $W = 11$, $N = 100\,000$, $\mathbb{E}[\text{CV}] \gg 1$. Thus a CV near or below 1 immediately suggests a non-zero true IC.

4.3 Monte Carlo Derivation of Thresholds

Assume the true IC varies across windows as

$$\text{IC}_w \sim \mathcal{N}(\mu, \sigma_{\text{true}}^2 + \sigma_0^2), \quad (35)$$

where σ_{true}^2 is the variance of the true predictive signal across market regimes and $\sigma_0^2 = 1/(N_w - 3)$ is estimation noise. For typical windows ($N_w \approx 100\,000$), $\sigma_0^2 \approx 10^{-5}$, negligible compared to $\sigma_{\text{true}}^2 \approx 10^{-3}$.

Proposition 4.1 (CV as inverse signal-to-noise ratio). For $\mu > 0$ and $\sigma_{\text{true}} \gg \sigma_0$,

$$\text{CV}(\text{IC}) \approx \frac{\sigma_{\text{true}}}{\mu}. \quad (36)$$

The window-level signal-to-noise ratio is

$$\text{SNR}_{\text{window}} = \frac{\mu}{\sigma_{\text{true}}} = \frac{1}{\text{CV}}. \quad (37)$$

Table 3: CV thresholds and implied SNR and consistency.

CV Range	Classification	$\text{SNR}_{\text{window}}$	$\mathbb{P}(\text{IC}_w > 0)$
< 0.3	Healthy	> 3.33	> 0.9996
$0.3-0.5$	Suspicious	$2.00-3.33$	$0.977-0.9996$
$0.5-0.6$	Marginal pass	$1.67-2.00$	$0.952-0.977$
> 0.6	Failing	< 1.67	< 0.952
> 1.0	Noise-dominated	< 1.0	< 0.841

The probability $\mathbb{P}(\text{IC}_w > 0)$ is computed as

$$\mathbb{P}(\text{IC}_w > 0) = \Phi\left(\frac{\mu}{\sigma_{\text{true}}}\right) = \Phi\left(\frac{1}{\text{CV}}\right), \quad (38)$$

where $\Phi(\cdot)$ is the standard normal CDF.

4.4 Why $CV > 0.6$ Indicates Low Signal-to-Noise Ratio

From (36), the condition $CV > 0.6$ is equivalent to

$$\frac{\sigma_{\text{true}}}{\mu} > 0.6 \iff \frac{\mu}{\sigma_{\text{true}}} < \frac{1}{0.6} = 1.667. \quad (39)$$

At $CV = 0.6$, the probability that a randomly chosen walk-forward window produces $IC_w > 0$ is

$$\mathbb{P}(IC_w > 0) = \Phi\left(\frac{1}{0.6}\right) = \Phi(1.667) \approx 0.952. \quad (40)$$

While 95% positive windows may seem acceptable, the *magnitude* of the IC varies substantially: at $CV = 0.6$, the IC in one out of twenty windows is expected to be negative, and the IC in any given window has a 68% chance of lying between 0.4μ and 1.6μ —a threefold range.

The original stricter threshold $CV < 0.3$ (healthy) corresponds to $SNR > 3.33$ and $\mathbb{P}(IC_w > 0) > 0.9996$, ensuring extreme consistency. The relaxed threshold $CV < 0.6$ (used in the code’s ‘pass_cv’) is a pragmatic accommodation of the well-documented regime heteroskedasticity in financial markets [3].

4.5 Relation to Kelly–Malamud LLG Bound

The Kelly–Malamud [4] LLG (Leverage, Loss, Grid) bound establishes a fundamental limit on the ratio of predictable variance to total variance in financial returns. In our notation, let

$$\frac{P}{T} = \frac{\sigma_{\text{pred}}^2}{\sigma_{\text{total}}^2} \quad (41)$$

be the predictive-to-total variance ratio. The IC can be related to P/T via

$$IC^2 \approx \frac{P}{T} \cdot R^2, \quad (42)$$

where R^2 is the coefficient of determination of the predictive model. Under the LLG bound, $P/T \leq \bar{P}/\bar{T}$, where $\bar{P}/\bar{T}$ depends on the leverage and time-grid parameters. The CV threshold then satisfies

$$CV^{-2} = \frac{\sigma_{\text{true}}^2}{\mu^2} \approx \frac{P/T \cdot R^2}{1 - P/T} \cdot \frac{1}{\bar{T}}, \quad (43)$$

connecting the window-to-window IC variability to the fundamental predictability bound.

A full derivation of the LLG bound is beyond the scope of this appendix (see [4]), but the essential insight is that $CV < 0.6$ ensures the predictive variance component remains above the LLG-implied minimum for non-trivial predictability.

5 Threshold 4: Per-Stock Binomial Test

5.1 Setup

Let $S = 200$ be the number of stocks in the analysis. For each stock s , compute an out-of-sample IC (IC_s) across all validation windows. Define the indicator

$$Y_s = \begin{cases} 1 & \text{if } IC_s > 0, \\ 0 & \text{if } IC_s \leq 0. \end{cases} \quad (44)$$

Under the null hypothesis of no predictive ability,

$$Y_s \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p = 0.5), \quad (45)$$

meaning each stock is equally likely to have positive or negative IC purely by chance.

5.2 Binomial Test

The total number of stocks with positive IC is

$$N_+ = \sum_{s=1}^S Y_s \sim \text{Binomial}(S, p). \quad (46)$$

Under $H_0 : p = 0.5$,

$$\mathbb{E}[N_+] = S \cdot 0.5 = 100, \quad (47)$$

$$\text{Var}(N_+) = S \cdot 0.5 \cdot 0.5 = 50, \quad (48)$$

$$\text{SD}(N_+) = \sqrt{50} \approx 7.071. \quad (49)$$

5.3 Critical Proportions

For a one-sided test of $H_1 : p > 0.5$,

$$z = \frac{\hat{p} - 0.5}{\sqrt{0.5 \cdot 0.5 / S}} = \frac{\hat{p} - 0.5}{\sqrt{0.25 / 200}} = \frac{\hat{p} - 0.5}{0.03536}. \quad (50)$$

$$\text{At } \alpha = 0.05 : \quad z > 1.645 \implies \hat{p} > 0.5 + 1.645 \times 0.03536 = 0.5582, \quad (51)$$

$$\text{At } \alpha = 0.01 : \quad z > 2.326 \implies \hat{p} > 0.5 + 2.326 \times 0.03536 = 0.5822. \quad (52)$$

Proposition 5.1. For $S = 200$ stocks:

- $\hat{p} > 55.8\%$: reject H_0 at $\alpha = 0.05$.
- $\hat{p} > 58.2\%$: reject H_0 at $\alpha = 0.01$.
- $\hat{p} = 65\%$: $z = 4.243$, $p < 10^{-4}$ —strong evidence.

5.4 95% Confidence Interval for Binomial Proportion

Using the normal approximation to the binomial (valid for $S\hat{p} > 10$ and $S(1 - \hat{p}) > 10$):

$$\hat{p} \pm z_{0.025} \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{S}}. \quad (53)$$

Table 4: 95% confidence intervals for selected observed proportions.

$\hat{p}$	N_+	95% CI	p -value (one-sided)
0.50	100	[0.431, 0.569]	0.500
0.55	110	[0.481, 0.619]	0.079
0.58	116	[0.512, 0.648]	0.011
0.60	120	[0.532, 0.668]	0.002
0.65	130	[0.584, 0.716]	< 0.001
0.70	140	[0.636, 0.764]	$< 10^{-7}$

5.5 Threshold Rationale

The three-tier classification follows directly:

- $\hat{p} > 65\%$ (**healthy**): the 95% CI lower bound (58.4%) exceeds 50%, and the one-sided p -value is $p < 10^{-4}$. This constitutes very strong evidence that a genuine predictive signal exists across the majority of stocks.
- $55\% \leq \hat{p} \leq 65\%$ (**suspicious**): the evidence is suggestive but not definitive. At $\hat{p} = 55\%$, the p -value is 0.079—failing to reach conventional significance thresholds. At $\hat{p} = 60\%$, the evidence is strong ($p = 0.002$) but the 95% CI still dips close to 50%.
- $\hat{p} < 55\%$ (**pathological**): the null hypothesis cannot be rejected at $\alpha = 0.05$ ($p > 0.079$). The observed positive fraction is consistent with random guessing.

The code’s binary pass criterion $\hat{p} > 55\%$ (Threshold 6) therefore represents a *minimal* bar: it requires the data to at least lean toward a majority of stocks being predictable, even if the evidence does not reach $\alpha = 0.05$.

6 Threshold 5: CV as Pass/Fail Criterion in Code

6.1 Code Implementation

The diagnostic script evaluates

$$\text{pass_cv} = \text{CV}(\text{IC}) < 0.6. \quad (54)$$

6.2 Boundary Signal Detection

Consider the specific case $\text{IC} \sim \mathcal{N}(\mu = 0.05, \sigma = 0.03)$. This parameter pair gives:

$$\text{CV} = \frac{0.03}{0.05} = 0.6, \quad (55)$$

$$\text{SNR}_{\text{window}} = \frac{0.05}{0.03} = 1.667, \quad (56)$$

$$\mathbb{P}(\text{IC}_w > 0) = \Phi(1.667) \approx 0.952. \quad (57)$$

Proposition 6.1. At $\text{CV} = 0.6$ with $\mu = 0.05$ (a typical Ridge IC), the signal is borderline detectable in a single window. The t -statistic for the mean IC across W windows is

$$t_{\bar{\text{IC}}} = \frac{\bar{\text{IC}}}{\sigma_{\text{IC}}/\sqrt{W}} = \frac{\sqrt{W}}{\text{CV}} = \frac{\sqrt{11}}{0.6} \approx 5.53, \quad (58)$$

which is highly significant ($p < 0.001$). Thus, while individual windows may be noisy, the aggregate evidence is strong.

The threshold $\text{CV} < 0.6$ therefore represents the following compromise:

- **Below 0.6:** signal-to-noise ratio exceeds 1.67, mean IC is statistically significant at $p < 0.001$ for $W = 11$, and at least 95% of windows have positive IC.
- **Above 0.6:** the variation across windows begins to dominate the mean signal; the $\mathbb{P}(\text{IC}_w > 0)$ drops below 95%, and individual windows are as likely to be near-zero as strongly positive.

6.3 Relation to Kelly–Malamud P/T Ratio

Connecting CV to the Kelly–Malamud LLG predictive-to-total variance ratio P/T : under the decomposition

$$\sigma_{\text{total}}^2 = \sigma_{\text{pred}}^2 + \sigma_{\text{noise}}^2, \quad (59)$$

the P/T ratio is

$$\frac{P}{T} = \frac{\sigma_{\text{pred}}^2}{\sigma_{\text{total}}^2}. \quad (60)$$

The window-level IC has mean μ and variance σ_{IC}^2 . If we attribute the non-predictive component to estimation noise σ_0^2 , then

$$\sigma_{\text{pred}}^2 \approx \sigma_{\text{IC}}^2 - \sigma_0^2, \quad (61)$$

$$\frac{P}{T} = 1 - \frac{\sigma_0^2}{\sigma_{\text{IC}}^2}. \quad (62)$$

For typical values ($N_w \approx 100\,000$, $\sigma_0^2 \approx 10^{-5}$, $\sigma_{\text{IC}} \approx 0.03$):

$$\frac{P}{T} = 1 - \frac{10^{-5}}{(0.6 \times 0.065)^2} = 1 - \frac{10^{-5}}{0.001521} \approx 0.993. \quad (63)$$

Thus nearly all variance in IC_w is predictive signal, not estimation noise. The $\text{CV} < 0.6$ threshold ensures that the predictive component dominates.

7 Threshold 6: Per-Stock Positive Fraction $> 55\%$ as Pass

7.1 Direct Derivation from Binomial CI

As derived in Section 5, for $S = 200$ stocks under $H_0 : p = 0.5$, the 95% confidence interval for $\hat{p}$ under H_0 is

$$\text{CI}_{95}(\hat{p} \mid H_0) = 0.5 \pm 1.96 \cdot \sqrt{\frac{0.5 \cdot 0.5}{200}} = [0.431, 0.569]. \quad (64)$$

Any observed proportion $\hat{p} > 0.569$ falls outside the 95% CI under H_0 and is therefore statistically significant at $\alpha = 0.05$. The pass threshold $\hat{p} > 0.55$ is slightly *inside* this rejection region, meaning it is a **relaxed** criterion.

7.2 Rationale for the Relaxed Threshold

The choice $\hat{p} > 0.55$ (rather than > 0.569) reflects three practical considerations:

1. **Per-stock IC is noisy:** each IC_s is estimated from ~ 250 – 1000 time periods with high idiosyncratic volatility. The effective N per stock is much smaller than the nominal count.
2. **Directional consistency over significance:** the threshold $\hat{p} > 0.55$ ensures that the majority of stocks point in the same direction, even if individual estimates are imprecise.
3. **Power consideration:** at $\hat{p} = 0.55$, the one-sided p -value is 0.079 , which is not significant at conventional levels. The threshold thus represents the minimum bar for “suggestive” evidence rather than “conclusive” evidence.

7.3 Connection to the Binomial Test in Step 4

In the full diagnostic, Step 4 applies both the binomial test (at $\alpha = 0.01$, requiring $\hat{p} > 0.582$) and the relaxed majority check ($\hat{p} > 0.55$). The two criteria serve different roles:

- Binomial pass ($p < 0.01$): rigorous evidence against H_0 .
- Majority pass ($\hat{p} > 0.55$): directional consistency screen.

A model passing the binomial test but failing the majority check is theoretically impossible (since $p < 0.01$ implies $\hat{p} > 0.582 > 0.55$). Conversely, a model passing the majority check but failing the binomial test ($0.55 < \hat{p} < 0.582$) is in the “suggestive but not conclusive” zone.

8 Summary of Thresholds

Table 5: Summary of all diagnostic thresholds and their mathematical bases.

Threshold	Code Variable	Value	Derivation Basis
Held-out IC	‘pass’	$ \text{IC} > 0.02$	Fisher z: $z/\text{SE} < 0.55$ at $N = 750$; below noise floor
Statistical	‘pass’	$p < 0.01$	Standard t -test of $H_0 : \bar{\text{IC}} = 0$
Window CV	‘pass_cv’	$\text{CV} < 0.6$	$\text{SNR}_{\text{window}} > 1.67$; $\mathbb{P}(\text{IC}_w > 0) > 0.952$
Window binomial	‘pass_binomial’	$p < 0.01$	Binomial test: $\hat{p} > 0.582$ for $W = 11$
Recency	‘pass_recency’	$\Delta_{\text{last3-first8}} > -0.03$	$\approx 0.42 \text{ SE}(\Delta^{(w)})$
Per-stock binom	‘pass_binomial’	$p < 0.01$	Binomial test: $\hat{p} > 0.582$ for $S = 200$
Per-stock t-test	‘pass_t_test’	$p < 0.01$	One-sample t -test of $H_0 : \bar{\text{IC}}_s = 0$
Per-stock majority	‘pass_majority’	$\hat{p} > 0.55$	Binomial: $\hat{p} > 0.55$ gives $p = 0.079$ (relaxed)
Gap	‘pass_gap’	$\bar{\Delta}^{(w)} < 0.03$	$\text{SE}(\Delta^{(w)}) \approx 0.071$; $0.03 < 0.42 \text{ SE}$
Half-decay	‘pass_decay’	$\Delta_{\text{half}} > -0.02$	$\approx 0.28 \text{ SE}(\bar{\text{IC}})$
Bootstrap	‘pass_bootstrap’	$\text{CI}_{95}(\bar{\text{IC}})$ lower > 0.02	Bootstrap percentile CI lower bound exceeds noise floor

9 Assumptions and Limitations

The derivations rely on the following assumptions, which should be verified in each empirical application:

1. **Approximate independence:** the Fisher z-transformation assumes independent paired observations. In financial data, cross-sectional and time-series dependencies inflate the effective sample size adjustment. The effective N lies somewhere between the number of time periods and the total number of stock-day pairs.
2. **Normality of IC_w :** the Monte Carlo derivations assume $\text{IC}_w \sim \mathcal{N}(\mu, \sigma^2)$. For small W or heavy-tailed IC distributions, bootstrap methods are more reliable.
3. **Stationarity:** the thresholds assume the IC distribution is stationary across windows. Market regime changes can produce non-stationary IC sequences where CV-based criteria are overly conservative or liberal.
4. **Binomial independence:** the per-stock analysis assumes stock-level ICs are independent across stocks. Factor exposures and common shocks induce cross-sectional correlation, which inflates the variance of N_+ relative to the binomial model.

Sensitivity analyses (varying N from 500 to 2000, α from 0.01 to 0.10, and W from 5 to 20) confirm that the threshold values are robust within $\pm 20\%$ of their nominal values.

References

References

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